

HCAHPS Patient Analysis Project

KPI Definition Document

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Tools Used

Power BI

Power Query

Excel

Submission Date

6 August 2026

1. Introduction

This document defines the Key Performance Indicators (KPIs) used in the HCAHPS Patient Analysis Dashboard. Each KPI includes its definition, DAX formula, and business importance. These KPIs help measure patient satisfaction, monitor healthcare quality, and support data-driven decision-making.

KPI 1 – National Top-box %

Definition

The National Top-box % represents the average percentage of patients who selected the most positive response for HCAHPS survey questions at the national level.

DAX Formula

National Top-box % = AVERAGE('national_results'[Top-box Percentage])

Business Importance

This KPI measures overall patient satisfaction across the country and helps healthcare organizations monitor healthcare quality and identify opportunities for improvement.

KPI 2 – National Middle-box %

Definition

The National Middle-box % represents the average percentage of patients who selected the neutral or middle response for HCAHPS survey questions at the national level.

DAX Formula

National Middle-box % = AVERAGE('national_results'[Middle-box Percentage])

Business Importance

This KPI helps healthcare organizations understand the proportion of patients with moderate satisfaction levels and identify areas where patient experience can be improved.

KPI 3 – National Bottom-box %

Definition

The National Bottom-box % represents the average percentage of patients who selected the least positive response for HCAHPS survey questions at the national level.

DAX Formula

National Bottom-box % = AVERAGE('national_results'[Bottom-box Percentage])

Business Importance

This KPI identifies areas of patient dissatisfaction and helps healthcare organizations prioritize quality improvement initiatives.

KPI 4 – State Average Top-box %

Definition

The State Average Top-box % represents the average percentage of patients who selected the most positive response for HCAHPS survey questions across all states.

DAX Formula

State Average Top-box % = AVERAGE('state_results'[Top-box Percentage])

Business Importance

This KPI helps compare patient satisfaction across different states and supports state-level healthcare performance evaluation.

KPI 5 – State Satisfaction Score

Definition

The State Satisfaction Score represents the average patient satisfaction level for each state based on the Top-box Percentage.

DAX Formula

State Satisfaction Score = AVERAGE('state_results'[Top-box Percentage])

Business Importance

This KPI enables comparison of patient satisfaction across states and helps identify high-performing and low-performing states.

KPI 6 – Regional Satisfaction Score

Definition

The Regional Satisfaction Score represents the average patient satisfaction level for each geographic region based on the Top-box Percentage.

DAX Formula

Regional Satisfaction Score =AVERAGE('state_results'[Top-box Percentage])

Business Importance

This KPI helps compare patient satisfaction across different regions and supports regional healthcare performance analysis.

KPI 7 – Survey Response Rate

Definition

The Survey Response Rate represents the average percentage of patients who responded to the HCAHPS survey.

DAX Formula

Survey Response Rate = AVERAGEX('responses', IFERROR(VALUE('responses'[Response Rate (%)]), BLANK()))

Business Importance

This KPI measures the participation level of patients in the survey. A higher response rate improves the reliability of patient satisfaction analysis.

KPI 8 – Number of Participating States

Definition

The Number of Participating States represents the total number of unique states included in the HCAHPS survey reporting.

DAX Formula

Number of Participating States =DISTINCTCOUNT('state_results'[State])

Business Importance

This KPI indicates the geographical coverage of the survey and helps understand the scope of the analysis.

KPI 9 – Number of Quality Measures

Definition

The Number of Quality Measures represents the total number of unique healthcare quality measures used to evaluate patient satisfaction.

DAX Formula

Number of Quality Measures = DISTINCTCOUNT('measures (2)'[Measure ID])

Business Importance

This KPI indicates the breadth of healthcare quality indicators included in the analysis.

KPI 10 – Highest Performing Measure

Definition

The patient experience measure with the highest National Top-box Percentage..

DAX Formula

Highest Performing Measure =TOPN(1, VALUES('measures (2)'[Measure]), [Average Top-box %],DESC)

Business Importance

Identifies the patient experience area where hospitals perform best, helping management recognize strengths and maintain high-quality service.

KPI 11 – Lowest Performing Measure

Definition

The patient experience measure with the lowest National Top-box Percentage.

DAX Formula

Lowest Performing Measure = TOPN(1, VALUES('measures (2)'[Measure]), [Average Top-box %], ASC)

Business Importance

Highlights the weakest-performing patient experience measure so improvement efforts can be focused on areas with the greatest need.

KPI 12 – Year-over-Year Satisfaction Change

Definition

Measures the change in Average Top-box % between the current reporting period and the previous reporting period.

DAX Formula

Year-over-Year Satisfaction Change = [Average Top-box %] - CALCULATE([Average Top-box %], PREVIOUSYEAR('reports'[Start Date]))

Business Importance

Helps management monitor whether patient satisfaction is improving or declining over time and evaluate the impact of quality improvement initiatives.

KPI 13 – Regional Ranking

Definition

Ranks each region based on its Regional Satisfaction Score.

DAX Formula

Region Ranking =RANKX(ALL('states'[Region]), [Regional Satisfaction Score], , DESC, DENSE)

Business Importance

Identifies the highest- and lowest-performing regions, helping healthcare administrators compare regional performance and allocate improvement efforts effectively.

KPI 14 – State Ranking

Definition

Ranks each state based on its State Satisfaction Score.

DAX Formula

State Ranking =IF(HASONEVALUE('states'[State]), RANKX(ALLSELECTED('states'[State]), [State Satisfaction Score], , DESC, DENSE), BLANK())

Business Importance

Provides a comparative ranking of states based on patient satisfaction, enabling organizations to identify top-performing states and focus improvement efforts on lower-performing states.